

B 1 1 2 0 3 0 4 $0^\circ, 360^\circ$ 5 $60^\circ, 300^\circ$ 6 180° 7 0.5
8 -0.5 9 0.5

C 1 1, 0.87, 0.71, 0.5, 0, -0.5 , -0.71 , -1 , -0.71 , 0, 1
2 2, 1.73, 1.41, 1, 0, -1 , -1.41 , -2 , -1.41 , 0, 2
3 1, 0.5, 0, -0.5 , -1 , -0.5 , 0, 1, 0, -1 , 1
4 -1 , -0.87 , -0.71 , -0.5 , 0, 0.5, 0.71, 1, 0.71, 0, -1
5 a $y = 2 \cos x$ b $y = \cos 2x$ c $y = \cos \frac{1}{2}x$
d $y = -\cos x$ 6 1 7 360° 8 3 9 180°

18 The graph of $y = \tan x^\circ$

A 1 0, 0.58, 1, 1.73, undefined, -1.73 , -1 , -0.58 , 0, 0.58,
1, 1.73, undefined, -1.73 , -1 , -0.58 , 0

B 1 continuous 2 $-270^\circ, -90^\circ, 90^\circ, 270^\circ$ 3 $180^\circ, 180^\circ$
4 amplitude 5 asymptotes 6 $-360^\circ, -180^\circ, 0^\circ, 180^\circ,$
 360° 7 $-315^\circ, -135^\circ, 45^\circ, 225^\circ$ 8 $-225^\circ, -45^\circ, 135^\circ,$
 315°

C 1 180° 2 $x = 90^\circ, x = 270^\circ$ 3 60° 4 $x = -30^\circ,$
 $x = 30^\circ, x = 90^\circ$ 5 0, 3, -3 , 0, 3, -3 , 0 6 0, 1,
undefined, -1 , 0, 1, undefined, -1 , 0 7 $y = \tan 2x$

19 Further trigonometric graphs

A 1 $a = 0.866, b = 90^\circ, c = 0.5, d = -0.5, e = 330^\circ, f = 0$
3 a $x = 45^\circ, 135^\circ$ b $x = 225^\circ, 315^\circ$ c $x = 210^\circ, 330^\circ$
d $x = 0^\circ, 180^\circ, 360^\circ$

B 1 a $90^\circ, 270^\circ$ b infinite c $107^\circ, 287^\circ$ d 215°
2 A($90^\circ, 3$), B($180^\circ, 0$), C($270^\circ, -3$), D($360^\circ, 0$) 3 $p = 3,$
 $q = 1, m = 2, n = \frac{1}{2}$ ($y = 3 \sin x$ and $y = 2 \cos \frac{1}{2}x$);
 $y = 3 \sin x$ has period 360° , amplitude = 3;
 $y = 2 \cos \frac{1}{2}x$ has period 720° , amplitude = 2

C 2 $\cos x$: 1, 0.5, 0, -0.5 , -1 , -0.5 , 0, 0.5, 1
 $\sec x$: 1, 2, undefined, -2 , -1 , -2 , undefined a 360°
b infinite c infinite
3 $\sin x$: 0, 0.5, 1, 0.5, 0, -0.5 , -1 , -0.5 , 0
 $\operatorname{cosec} x$: undefined, 2, 1, 2, undefined, -2 , -1 , -2 ,
undefined a 360° b infinite c infinite

20 Simple trigonometric equations

A 1 37° 2 67° 3 22° 4 25° 5 73° 6 41° 7 57°
8 30° 9 39° 10 58° 11 67° 12 44° 13 48° 14 51°
15 57° 16 89° 17 73° 18 84° 19 69° 20 6° 21 67°
22 62° 23 32° 24 74° 25 28° 26 25° 27 19° 28 29°

B 1 143° 2 166° 3 121° 4 175° 5 102° 6 115°
7 139° 8 99° 9 153° 10 129° 11 117° 12 108°
13 $30^\circ, 150^\circ$ 14 $44^\circ, 136^\circ$ 15 $59^\circ, 121^\circ$ 16 $13^\circ, 167^\circ$
17 65° 18 120° 19 136° 20 30° 21 27° 22 124°
23 68° 24 126° 25 $24^\circ, 156^\circ$ 26 60° 27 45° 28 $53^\circ,$
 127° 29 107° 30 117° 31 $30^\circ, 150^\circ$ 32 72° 33 48°

34 112° 35 120° 36 $49^\circ, 131^\circ$

C 1 $60^\circ, 300^\circ$ 2 $210^\circ, 330^\circ$ 3 $56^\circ, 236^\circ$ 4 $135^\circ, 315^\circ$
5 $117^\circ, 297^\circ$ 6 $229^\circ, 311^\circ$ 7 $58^\circ, 238^\circ$ 8 $201^\circ, 338^\circ$
9 $38^\circ, 322^\circ$ 10 $120^\circ, 300^\circ$ 11 $108^\circ, 208^\circ$ 12 $233^\circ,$
 307° 13 90° 14 $63^\circ, 243^\circ$ 15 180° 16 $41^\circ, 319^\circ$
17 270° 18 $50^\circ, 230^\circ$ 19 $42^\circ, 138^\circ$ 20 $27^\circ, 207^\circ$
21 $71^\circ, 289^\circ$ 22 $127^\circ, 307^\circ$ 23 $30^\circ, 150^\circ$ 24 $124^\circ,$
 304°

21 Trigonometric relationships

A 1 true 2 true 3 true 4 true 5 true 6 true 7 true
8 true

22 Review: trigonometry

A 1 9.9 cm 2 6.6 cm 3 33.9 cm 4 $44^\circ 54'$ 5 $34^\circ 23'$
6 $54^\circ 19'$ 7 31.9 cm 8 31.3 cm 9 31.1 cm 10 82°
11 61° 12 40° 13 136 cm^2 14 321 cm^2 15 234 cm^2

B 1 100° 2 103° 3 118° 4 3.8 m 5 4.7 m 6 1.9 m
7 $91^\circ 28'$ 8 $58^\circ 16'$ 9 $93^\circ 12'$ 10 902 cm^2 11 401 m
12 217 mm 13 347.4 m 14 6924 mm
15 AB = 56.0 m, BD = 37.5 m

C 1 23.9 cm 2 27.6 cm 3 19.3 cm 4 40.5 cm 5 $45^\circ 11'$
6 $54^\circ 31'$ 7 $35^\circ 5'$ 8 $116^\circ 48'$ 9 245 mm 10 $77^\circ 3'$
11 186.7 m 12 perimeter = 298 m, area = 5404 m^2
13 perimeter = 348 m, area = 7142 m^2